Psychoneuroendocrinology, Vol. 8, No. 4, pp. 401 – 409, 1983.
Printed in Great Britain.

0306 – 4530/83 $3.00 + 0.00
© 1983 Pergamon Press Ltd.

BEHAVIORAL, BUT NOT PHYSIOLOGICAL, ADAPTATION TO REPEATED SEPARATION IN MOTHER AND INFANT PRIMATES

CHRISTOPHER L. COE, JEFFREY C. GLASS, SANDRA G. WIENER and SEYMOUR LEVINE
Department of Psychiatry and Behavioral Sciences, Stanford University School of Medicine, Stanford, CA 94305, U.S.A.

(*Received* 15 *February* 1983)

SUMMARY

Mother and infant squirrel monkeys were subjected to a series of brief separations in order to evaluate how behavioral and physiological responses change following multiple exposures to stress. Beginning when the infants reached three months of age, their behavioral and hormonal responses were assessed during six 1-hr separations; an additional five dyads served as controls for the effect of repeated disturbance. The separated infants showed a marked and progressive decrease in distress calling across time, but no change was observed in the high levels of agitated activity or the plasma cortisol response to separation. This finding questions the traditional use of distress vocalizations as a measure of stress and indicates that certain types of behavior can change independently of physiological arousal responses, which may continue to occur even after repeated exposures to stress.

Key Words—Squirrel monkey; stress; cortisol; development; separation; habituation.

IN SPITE OF extensive research on the effects of stress, it still is difficult to predict how stressful experiences will affect an individual's subsequent response to stimuli of an aversive nature. Even in the simplest case, when the same stressor reoccurs, repeated exposure can lead to habituation under certain circumstances and sensitization under others (Hennessey & Levine, 1979; File, 1982). Most theories have emphasized the importance of cognitive factors, such as those related to predictability and control, for determining the consequences of stressful experiences (Lazarus, 1968; Seligman, 1975; Levine, 1980; Rodin, 1980). In general, greater predictability and a sense that one's actions are efficacious in dealing with the stressful situation usually lead to a reduction in stress, both at the subjective and physiological levels. However, additional theoretical approaches also are required to explain why physiological and behavioral parameters often adapt at different rates, and sometimes in different directions, during repeated or prolonged stress (Mason, 1968; Ursin *et al.*, 1978; Rose, 1980).

Although they originally proposed it in another context, Groves and Thompson (1970) offered another perspective that is useful for predicting some of the effects of repeated exposure to stress. They suggested that skeletal – motor systems should be expected to habituate following repeated stimulation, whereas arousal systems would be more likely to remain constant or to be sensitized. Independent support for this proposition has been generated from stress research showing that behavioral responses typically subside following repeated exposure to novelty or electric shock, while physiological measures of arousal, such as adrenal activation, are less likely to change (Natelson *et al.*, 1976; Hennessy & Levine, 1979). For example, rats showed no change in their corticosterone

response after 10 daily exposures to novelty, and actually showed an enhanced corticosterone response after 10 daily exposures to shock (Hennessy *et al.*, 1977). Although it is possible to reduce the adrenocortical response after a sufficient number or duration of exposures (Mikulaj & Mitro, 1973), other types of endocrine activation, such as increased epinephrine secretion, often continue to occur (Kvetnansky & Mikulaj, 1970; Hansen *et al.*, 1980).

Further resolution of the factors underlying this apparent dissociation between behavioral and physiological responses is of obvious importance, both for methodological and theoretical reasons, and there is a real need to test these hypotheses in situations that involve more complex types of psychological stress. In the following research, we attempted to apply these concepts by evaluating how monkeys reacted to repeated disruption of the emotional bond between mother and infant. The traumatic consequences of mother–infant separation have been the focus of extensive study in humans (Bowlby, 1969; Weinraub & Lewis, 1977) and in a number of animal species (Mineka & Suomi, 1978; Hofer, 1981; Rosenblum & Plimpton, 1981). Involuntary separation of mother and infant usually results in severe behavioral disturbance and, at least in the case of nonhuman primates, activation of the endocrine and autonomic responses that typically occur following other physical and psychological stressors (Reite *et al.*, 1978; Coe *et al.*, 1983). With regard to the effect of repeated separation experiences, the general view has been that the infant's behavioral response to separation remains constant or worsens (Bowlby, 1969; Suomi *et al.*, 1970; Mineka, 1982). However, this view has been based primarily on the long-term reaction to prolonged and traumatic separations; more recently there has been renewed interest in the recurrent finding of a marked diminution in "protest" behavior during the first day or across separations (Rosenblum & Plimpton, 1981; Suomi *et al.*, in press). Since this type of behavioral transition is germane to the issue of how infants cope with repeated stress, we decided to evaluate the behavioral and adrenocortical response of mother–infant dyads across a series of brief separations. The separation conditions were kept constant and relatively benign, allowing us to assess how behavioral and hormonal changes reflected psychological changes across six separations that were spaced at 9 to 11 day intervals.

METHODS

Subjects

The subjects of this study were 10 mother–infant pairs of squirrel monkeys (*Saimiri sciureus*). The mothers had been imported as adults from Guyana and were well acclimated to the laboratory. The infants were born into small maternal groups and were undisturbed prior to the first separation that occurred at three months of age. The subjects lived in three small maternal groups consisting of either three or four dyads.

Housing

The three groups were housed in identical wire mesh pens (1.2 × 1.2 × 1.6 m) that were located in a large room of a laboratory building with regulated environmental conditions. Each pen contained three metal perches. Visual contact between pens was prevented by solid partitions. Wayne Monkey Diet and water were available *ad lib*; fresh fruit and vitamin supplements were provided twice each week.

Procedure

Five of the mother–infant pairs were assigned to the separation condition, while the remaining five dyads served as control subjects, designated basal. The separated subjects underwent a series of six 1-hr separations that were conducted at 9 to 11 day intervals beginning when the infants reached three months of age. These

separations were carried out by removing an infant from its mother at 0900 hr and placing it in a small cage in another room. During the separation the infant did not have visual or auditory contact with other monkeys. At the end of the separation it was immediately returned to its mother in the home cage. The mother – infant dyads assigned to the basal condition also lived in these social groups, but they were not separated, in order to control for any nonspecific effects of the blood sampling procedures.

During the last 30 min of each separation, the behavioral responses of the infants were observed and then a blood sample was obtained from both members of the dyad before returning the infant to the social group. Blood samples were also obtained from basal subjects at comparable 9 to 11 day intervals, although these samples were collected prior to any other manipulation on that day. The blood sampling procedure involved rapid anesthetization with ether and the collection of a 0.4 ml sample via cardiac puncture. This technique has proved invaluable, since it was possible to routinely collect samples within 2 min of capture and thereby to minimize the effects of any rapid hormone changes due to the disturbance of handling. The heparinized blood was centrifuged for 20 min at 2000 rpm, the plasma frozen, and the radioimmunoassay method of Klemm & Gupta (1975) used to determine plasma cortisol levels. Further details of the blood sampling and radioimmunoassay procedures are available in Coe *et al.* (1978).

Behavioral analysis

The behavioral observations focused exclusively on the infants' responses, because prior studies had indicated that mothers showed a limited and relatively invariant behavioral reaction to this type of test situation (Coe *et al.*, 1983; Suomi *et al.*, 1983). Observations of the separated infants were conducted through one-way glass and recorded on a checksheet. Although a number of behaviors were scored, the two most pertinent behaviors were distress vocalizations and agitated activity. Thus, the total frequency of the infants' distress calls (i.e. the species-specific isolation peep) and the percent of time spent active (at 30-sec intervals) have been used in the presentation of results. In addition, on the days intervening between separations, the infants' undisturbed behavior was observed in the social group. These observations were used to ascertain the effect of the separation experience on normative behavior and consisted of six 30-min observation sessions that focused on mother – infant relations and infant interactions with others. Both behavioral and hormonal data have been evaluated with repeated measures analyses of variance, and post hoc comparisons have been made with the Newman – Keuls test.

RESULTS

Figure 1 indicates that the repeated separations caused marked adrenal activation in both mothers and infants. Plasma cortisol levels in separated mothers averaged 244 μg%, which was a significant elevation over the typical morning levels of 152 μg% obtained in basal mothers ($F = 18.89$, $df = 1,8$; $p < 0.002$). Similarly, separated infants had an average cortisol level of 354 μg%, which was dramatically higher than the average level of

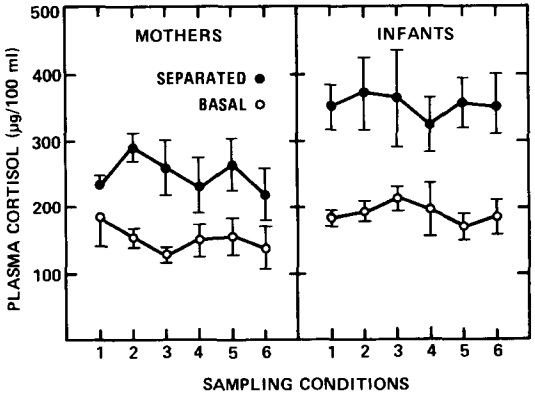


FIG. 1. The adrenocortical response of mother and infant monkeys to repeated blood sampling and to six 1-hr separations.

192 μg% in the undisturbed infants ($F = 21.59$, $df = 1,8$; $p < 0.002$). There was no evidence of any change in this adrenal response to separation across the six manipulations, nor did repeated blood sampling alter basal cortisol levels in either mothers or infants.

In spite of this constant physiological response, the infants' behavioral responses changed dramatically across separations. Infants responded with distress vocalizations and agitated activity during each separation, but the calling response decreased significantly with each succeeding separation, ($F = 13.38$, $df = 5,20$; $p < 0.001$) (Figure 2). The frequency of calling decreased stepwise from the first through the third separations ($p < 0.05$) and then remained at comparatively low levels during the fourth to sixth separations. This overall decline reflected a high rate of call emission during the first observation, a within-session decline during the second and third separations ($p < 0.01$), and a more constant and lower rate of calling during the fourth through sixth separations. It is of interest that this pattern of decrease occurred in infants that typically showed high levels of calling as well as in low-level callers, although these individual differences in calling rate did not correlate with the cortisol measures. Similarly, calling did not correlate with the infants' level of agitated activity, which remained high during all of the separations.

Despite the clear indications of distress during the separations, these manipulations did not have long-term effects on the undisturbed behavior of either the basal or the separated subjects. Evaluation of mother – infant contact, approach and retrieval behavior, as well as infant play and exploratory behavior on the days between separations did not reveal any significant differences between separated and basal subjects. In keeping with our

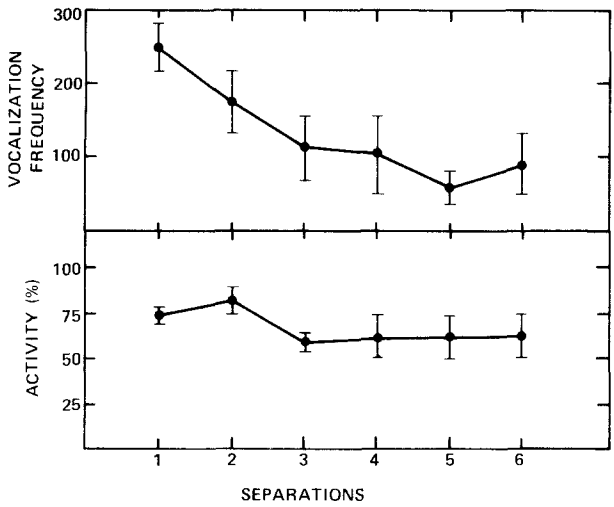


FIG. 2. The behavioral response of squirrel monkey infants during six repeated separations. Total number of isolation peeps and percent of time spent moving during the 30-min observations are portrayed.

expectations, there was a significant developmental decline in mother–infant contact ($F = 3.35$, $df = 5,40$; $p < 0.02$), a significant increase in time spent far away from the mother ($F = 6.61$, $df = 5,40$; $p < 0.001$), and a significant increase in infant play ($F = 3.02$, $df = 5,40$; $p < 0.002$) across time, but the six brief separation experiences did not alter the normal rate of behavioral maturation.

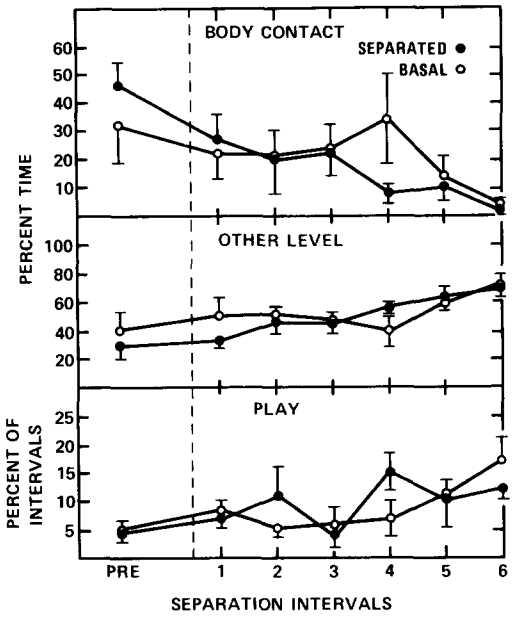


FIG. 3. Normative behavior of basal and separated subjects observed on days between manipulations. The figures illustrate the time spent by each infant in contact with its mother (upper), maximally separated from her (middle), and in social play with other infants (lower).

To ensure that the decrease in distress calling observed during the separation was not simply a function of maturation, older infants were separated for the first time at six months of age: several weeks after their sixth blood sample, the basal infants were subjected to a single separation. They responded with a high rate of 150 calls per 30 min, even though they were relatively independent of their mothers at this age. These older infants also showed the typical cortisol response to separation, averaging 380 μg%, which was significantly higher than their last basal value [$t(8) = 4.45$, $p < 0.05$].

DISCUSSION

The results from this experiment confirm previous studies which have shown that separation of mother and infant squirrel monkeys results in marked activation of the pituitary–adrenal system (Mendoza *et al.*, 1978; Coe & Levine, 1981). Plasma cortisol increases also have been observed in other primate species (Hill *et al.*, 1973; Gunnar *et al.*, 1981), but the high levels of adrenal corticosteroids that normally occur in the squirrel monkey permit a particularly dramatic demonstration of this response. Moreover, based

on our current data, it appears that neither previous separation experiences nor the infant's age alters the adrenal response to brief separations.

The equivalent cortisol response across the six separations is particularly interesting when considered in light of the rapidly changing behavioral response. As reported in a study on rhesus monkeys (Suomi *et al.*, 1983), the frequency of the infants' distress vocalizations decreased both within and across sessions. In our case, the data bore striking resemblance to an extinction curve: during the second and third separations there was a significant within-session decline in calling, and calling was maintained thereafter at a relatively constant lower rate. In another experiment we have shown that, when a separation is prolonged for 6 to 24 hr, a progressive decrease in calling also occurs during the first separation experience (Coe *et al.*, 1983). On the basis of the vocalization data alone, one might have concluded that the infants had undergone a rapid psychological adaptation and were less emotionally distressed. However, since the agitated activity and adrenal activation continued to occur, a more parsimonious interpretation is that the decrease in calling was due to its ineffectiveness in eliciting the appropriate reinforcer, the mother. In this interpretation, the infants' vocalizations should be viewed as signals that are normally used to evoke proximity-seeking and caregiving behavior, and one would predict that the rate of call emission should follow a predictable pattern depending on reinforcement contingencies of the separation environment. Indeed, in other separation studies we have found that calling rate is extremely stimulus-dependent. Infant calling is significantly higher, for example, when the mother is visually present, but inaccessible, than when the infant is totally isolated (Coe *et al.*, 1983). In this latter study, four of eight infants had actually extinguished their calling by 6 hr in the absence of the appropriate reinforcer, their mother, whereas all eight infants were still calling after 24 hr in the presence of their mother.

The susceptibility of calling to environmental influence and its capacity to vary independently of physiological state raise a number of issues which warrant brief discussion. First, the common use of vocalization rate as the primary or sole index of distress, as has been done in many separation studies, is probably inappropriate. While calling is a predictable and important dimension of the separation response, the duration and rate of calling appear to be too dependent upon external factors to serve as a reliable measure of internal state. The second issue of concern is the capacity of behavioral and physiological measures to vary independently of one another. In keeping with the prediction of Groves and Thompson (1970), vocalization changed rapidly during repeated exposure, whereas cortisol, our physiological measure of arousal, did not. Thus, as reported in studies on other types of stressors, such as novelty and electric shock (Ader, 1969; Natelson *et al.*, 1976; Hennessy & Levine, 1979), we can generalize that there will be a quicker behavioral adjustment to noxious stimuli and that physiological activation will subside only as behavioral and psychological processes facilitate coping with the aversive situation.

Finally, it is necessary that we try to reconcile our finding of behavioral adaptation with the general view that the infant's behavioral response either remains the same or worsens across separation (Bowlby, 1969; Suomi *et al.*, 1970; Spencer-Booth & Hinde, 1971; Mineka *et al.*, 1981). In part, the discrepancy in conclusions is due to a greater emphasis in

previous studies on the protracted response to separation over several days. Thus, a decline in "protest" behavior and a lapse into self-directed activities and social withdrawal—the so-called "depressive phase"—was considered to be a more severe reaction. However, we also would argue that any generalization from these earlier studies on the incidence of depression during repeated separation should be carefully qualified. Recent reviews have indicated that the occurrence of depressive behavior in primate infants is actually quite dependent on the separation environment—occurring primarily in those situations where a sense of helplessness has been evoked by imposing additional demands on the infant, and only in certain individuals that are particularly sensitive to stress due to genetic propensities or early rearing experiences (Rosenblum & Plimpton, 1981; Mineka, 1982; Suomi *et al.*, in press). With regard to the latter issue, many of the repeated separation studies have been conducted on infants reared without mothers (i.e. peer separations), and these monkeys are more vulnerable to depression than are mother-reared infants (Kraemer & McKinney 1979; Suomi *et al.*, in press). Similarly, even in those primate species which do not normally show depressive behavior, such as the bonnet macaque, it is possible to evoke depression by rearing them in highly stressful environments (Rosenblum & Plimpton, 1981) or by leaving them in threatening situations during the separation (Kaufman & Stynes, 1978).

Another important procedural variable that may have accounted for the more severe behavioral response across separations in previous studies was that the infants were reunited after separation either in disrupted social environments or with other traumatized subjects. Research on children and nonhuman primates indicates that the long-term consequences of separation are determined in large part by the quality of the reunion environment (Bowlby, 1969; McGinnis, 1980). Separation experiences are the most debilitating when the reunion environment is not supportive, and the infant may even show a severe secondary reaction when the mother is unresponsive (Kaufman & Rosenblum, 1967; Bowlby, 1969). In contrast to the cumulative and worsening reaction described in previous reports, in our study, where a stable reunion environment was available, we found that the infants' behavioral maturation was not impaired by the trauma of brief separations. Therefore, we would portray the primate infant as an extremely resilient organism that actively attempts to cope with stress, including that which is evoked by repeated experiences with maternal loss.

This research was supported by grants MH-23645 from NIMH, HD-02881 from NICHHD, and Research Scientist Award MH-19936 to S. Levine.

REFERENCES

ADER, R. (1969) Adrenocortical function and the measurement of emotionality. *Ann. N.Y. Acad. Sci.* **159**, 791 – 805.

BOWLBY, J. (1969) *Attachment and Loss: Attachment,* Vol. 1. Basic Books, New York.

COE, C. L. & LEVINE, S. (1981) Normal responses to mother-infant separation in nonhuman primates. In *Anxiety: New Research and Changing Concepts,* D. F. Klein and J. G. Rabkin (Eds.), pp. 155 – 177. Raven Press, New York.

COE, C. L., MENDOZA, S. P., DAVIDSON, J. M., SMITH, E. R., DALLMAN, M. F. & LEVINE, S. (1978) Hormonal response to stress in the squirrel monkey *(Saimiri sciureus). Neuroendocrinology* **26**, 367 – 377.

COE, C. L., WIENER, S. G. & LEVINE, S. (1983) Psychoendocrine responses of mother and infant monkeys to disturbance and separation. In *Symbiosis in Parent–Young Interactions,* L. A. Rosenblum and H. Moltz (Eds.). Plenum Press, New York (in press).

FILE, S. E. (1982) The rat corticosterone response: habituation and modification by chlordiazepoxide. *Physiol. Behav.* **29,** 91–96.

GROVES, P. M. & THOMPSON, R. F. (1970) Habituation: a dual process theory. *Psychol. Rev.* **77,** 419–450.

GUNNAR, M. R., GONZALEZ, C. A., GOODLIN, B. L. & LEVINE, S. (1981) Behavioral and pituitary–adrenal responses during a prolonged separation period in infant rhesus macaques. *Psychoneuroendocrinology* **6,** 65–75.

HANSEN, J. R., STOA, K. F., SCHYTTEBLIX, A. & URSIN, H. (1980) Urinary levels of epinephrine and norepinephrine in parachutist trainees. In *Psychobiology of Stress,* H. Ursin, E. Baade and S. Levine (Eds.), pp. 63–74. Academic Press, New York.

HENNESSY, J. W., LEVIN, R. & LEVINE, S. (1977) Influence of experiential factors and gonadal hormones on pituitary-adrenal response of the mouse to novelty and electric shock. *J. comp. Physiol. Psychol.* **91,** 770–777.

HENNESSY, J. W. & LEVINE, S. (1979) Stress, arousal, and the pituitary–adrenal system: a psychoendocrine hypothesis. In *Progress in Psychobiology and Physiological Psychology,* J. M. Sprague and A. N. Epstein (Eds.), Vol. 8, pp. 133–178. Academic Press, New York.

HILL, S. D., MCCORMACK, S. A. & MASON, W. A. (1973) Effects of artificial mothers and visual experience on adrenal responsiveness of infant monkeys. *Devl Psychobiol.* **6,** 421–429.

HOFER, M. A. (1981) Parental contributions to the development of their offspring. In *Parental Care in Mammals,* D. J. Gubernick and P. H. Klopfer (Eds.), pp. 77–116. Plenum Press, New York.

KAUFMAN, I. C., & ROSENBLUM, L. A. (1967) The reaction to separation in infant monkeys: anaclitic depression and conservation withdrawal. *Psychosom. Med.* **29,** 648–675.

KAUFMAN, I. C. & STYNES, A. J. (1978) Depression can be induced in a bonnet macaque infant. *Psychosom. Med.* **40,** 71–75.

KLEMM, W. & GUPTA, D. (1975) A routine method for the radioimmunoassay of plasma cortisol without chromatography. In *Radioimmunoassay of Steroid Hormones,* D. Gupta (Ed.), pp. 143–151. Verlag Chemie, Weinheim, Germany.

KRAEMER, G. W. & MCKINNEY, W. T. (1979) Interactions of pharmacological agents which alter biogenic amine metabolism and depression. *J. Affect. Disorders* **1,** 33–54.

KVETNANSKY, R. & MIKULAJ, L. (1970) Adrenal and urinary catecholamines during adaptation to repeated immobilization stress. *Endocrinology* **87,** 738–743.

LAZARUS, R. S. (1968) Emotions and adaptation: conceptual and empirical relations. *Nebr. Symp. Motiv.* **16,** 175–266.

LEVINE, S. (1980) A coping model of mother–infant relationships. In *Coping and Health,* S. Levine and H. Ursin (Eds.), pp. 87–99. Plenum Press, New York.

MASON, J. W. (1968) A review of psychoendocrine research on the pituitary–adrenal cortical system. *Psychosom. Med.* **30,** 576–607.

MCGINNIS, L. M. (1980) Maternal separation studies in children and nonhuman primates. In *Maternal Influences and Early Behavior,* R. W. Bell and W. P. Smotherman (Eds.), pp. 311–336. Spectrum, New York.

MENDOZA, S. P., SMOTHERMAN, W. P., MINER, M. T., KAPLAN, J. & LEVINE, S. (1978) Pituitary–adrenal response to separation in mother and infant squirrel monkeys. *Devl Psychobiol.* **11,** 169–175.

MIKULAJ, L. & MITRO, A. (1973) Endocrine functions during adaptation to stress. *Adv. exp. Med. Biol.* **33,** 631–638.

MINEKA, S. (1982) Depression and helplessness in primates. In *Primate Behavior and Child Nurturance,* H. Fitzgerald, J. Mullins and P. Gage (Eds.), Vol. 3, pp. 197–242. Plenum Press, New York.

MINEKA, S. & SUOMI, S. J. (1978) Social separation in monkeys. *Psychol. Bull.* **85,** 1376–1400.

MINEKA, S., SUOMI, S. J., & DELIZIO, R. (1981) Multiple separations in adolescent monkeys: an opponent process interpretation. *J. exp. Psych. Gen.* **110,** 56–85.

NATELSON, B. H., KRASNEGOR, N. & HOLADAY, J. W. (1976) Relations between behavioral arousal and plasma cortisol levels in monkeys performing repeated free-operant avoidance sessions. *J. comp. Physiol. Psychol.* **90,** 958–969.

REITE, M., SHORT, R., KAUFMAN, I. C., STYNES, A. J. & PAULEY, J. D. (1978) Heart rate and body temperature in separated monkey infants. *Biol. Psychiat.* **13,** 91–105.

RODIN, J. (1980) Managing the stress of aging: the role of control and coping. In *Coping and Health,* S. Levine and H. Ursin (Eds.), pp. 171–202. Plenum Press, New York.

ROSE, R. M. (1980) Endocrine responses to stressful psychological events. *Psychiat. Clin. N.A.* **3,** 251–275.

ROSENBLUM, L. A. & PLIMPTON, E. H. (1981) Adaptation to separation: the infant's effort to cope with an altered environment. In *The Common Child: Genesis of Behavior,* M. Lewis and L. A. Rosenblum (Eds.), Vol. 3, pp. 225 – 257. Plenum Press, New York.

SELIGMAN, M. E. P. (1975) *Learned Helplessness: On Depression, Development and Death.* W. H. Freeman, San Francisco.

SPENCER-BOOTH, Y. & HINDE, R. A. (1971) Effects of 6 days separation from mother in 18- to 32-week-old rhesus monkeys. *Anim. Behav.* **19,** 174 – 191.

SUOMI, S., HARLOW, H. & DOMEK, C. (1970) Effect of repetitive infant – infant separation of young monkeys. *J. Abnorm. Psychol.* **76,** 161 – 172.

SUOMI, S. J., MINEKA, S. & DELIZIO, R. D. Short- and long-term effects of repetitive mother – infant separations on social development in rhesus monkeys. *Devl Psychol.* (in press).

URSIN, H., BAADE, E. & LEVINE, S. (1978) *Psychobiology of Stress: A Study of Coping Men.* Academic Press, New York.

WEINRAUB, M. & LEWIS, M. (1977) The determinants of children's responses to separation. *Monogr. Soc. Res. Child Devl* **42,** Serial No. 172.